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Roll No

EE/EX-223-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Analog Electronics

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions out of eight questions.
ii) All questions carry equal marks.
iii) Assume suitable data, if required.

1. a) Write some of the difference between p-n junction and p-n junction diode? Explain with a suitable example.
b) Explain the temperature dependence and break down characteristics of p-n junction diode.
2. a) Discuss the comparison between varacter diode, Schottky diode and Tunnel Diode.
b) What do you mean by Transistors? Give a brief classification of all types of transistors.
3. a) Explain the comparison between CB, CE and CC configuration.
b) Derive a relation for drain current I_d for E MOSFET.
4. a) Explain the concept of feedback in any circuit? Explain it with example.
b) Define and explain the Barkhausen Criterion. Explain the concept of frequency oscillations.

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5.
 - a) Discuss about the current series and current shunt feedback with suitable circuit.
 - b) Write any five differences between the push pull and complementary push pull amplifiers.
6.
 - a) Write short note on De rating curve. Explain the class AB amplifier. Derive its relation as well.
 - b) Write short note on Hartley colpitts oscillator and UJT Oscillator circuits.
7.
 - a) Derive expressions for the multi vibrators and square wave generators for the circuits which behave as an OPAMPS.
 - b) Derive an expression for the Op-Amp inverting and non-inverting circuits.
8. Write short notes on any four of the following:
 - a) Schmitt Trigger
 - b) Wein Bridge Oscillator
 - c) LED
 - d) FET
 - e) Cascade amplifier
 - f) Boot strapping technique
